

Baby universe metric equivalent to an interior black-hole metric

Pedro F. González-Díaz

Instituto de Óptica "Daza de Valdés", CSIC, Serrano 121, E-28006 Madrid, Spain

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It is shown that the maximally extended metric corresponding to a large wormhole is the unique possible wormhole metric whose baby universe sector is conformally equivalent to the maximal inextendible Kruskal metric corresponding to the interior region of a Schwarzschild black hole whose gravitational radius is half the wormhole neck radius. The physical implications of this result in the black hole evaporation process are discussed.

The Hawking effect [1] by which an external observer detects thermal radiation coming from a black hole is just a semiclassical process. The approximation is only justified when the mass of the black hole is much greater than the Planck mass. What happens in the latest stages of the evaporation process is not quite clear, however [2]. A reasonable conjecture could be that black holes disappear completely, taking with them all the particles that fell in to form the black hole and the antiparticles to emit radiation. Hawking has conjectured [3] that all these particles disappear by going off through a wormhole into a little closed universe, which branches off from our region of spacetime. According to Hawking [3], the other end of the wormhole would appear to be another black hole which formed from the collapse of a star made up of massive antifermions, and evaporated, giving off the antiparticles to the particles forming the thermal radiation emitted by the first black hole in our region.

Thus, starting from an initial state which is just flat space, the final state of the black hole evaporation process will be flat space plus a number of baby closed universes. Such a topology change is only allowed when one is dealing with a spacetime manifold provided with a positive definite metric, where the four directions of spacetime are considered at the same footing, i.e. for a euclidean manifold with imaginary time.

Difficulties with the above guess appear when one tries to implement it in a four-dimensional manifold.

If we would consider any classical lorentzian manifold, the Geroch theorem [4] simply forbids any topology change to occur. Since the spacetime structures of four-dimensional euclidean black holes and wormholes are considered to be different, it is not quite clear how a black hole spacetime could be eventually converted into a wormhole spacetime, or vice versa, at a given stage of their evolution, even though they are not classical lorentzian entities provided with a globally hyperbolic metric.

For microscopic black holes, this transition could still be justified under the claim that the spacetime geometry does not have an objective existence, but has become an essentially observer-dependent quantum construct. However, it is very difficult to believe, in principle, that such a justification might be kept for macroscopic black holes where the spacetime geometry ought to be sufficiently well defined for black hole and wormhole geometries to be distinguishable.

On the other hand, according to current ideas about wormholes [5] (see, however, ref. [6]), it appears that the probability for a macroscopic wormhole (i.e., a wormhole whose throat size is of the order of the solar radius) to exist in the whole universe is overwhelmingly close to zero. Therefore, if in the case of handles of macroscopic size, it is possible for a closed universe the size of a macroscopic black hole to branch off, and one would interpret [3] such a branching as a black hole that formed and evapo-

rated again, the occurrence of such black holes will appear to be highly suppressed.

We are going to show nevertheless that there is a given kind of wormhole metric whose maximal analytical extension is essentially equivalent to the maximally extended black hole metric for the region beyond the event horizon of the black hole, solving in this way the above mentioned difficulties with the Hawking conjecture.

A four-dimensional black hole instanton corresponds to the spherically symmetric solution to euclidean *pure* gravity, with metric

$$ds_{\text{bh}}^2 = \left(1 - \frac{2m}{r}\right) d\tau^2 + \left(1 - \frac{2m}{r}\right)^{-1} dr^2 + r^2 d\Omega_2^2, \quad (1)$$

where τ denotes euclidean time, $\tau = -it$, and $d\Omega_2^2$ is the metric on the unit two-sphere. Metric (1) is euclidean only for $r > 2m$. It should be expected that the maximal analytical extension of this metric into the region $r < 2m$ would be everywhere euclidean.

Three different wormhole instantons are now known. They come about as solutions of the euclidean Einstein equations for gravity coupled to special kinds of *matter* fields in given particular ways. The available four-dimensional spherically symmetric wormhole metrics,

$$ds_{\text{wm}}^2 = \left(1 - \frac{M^n}{r^n}\right)^{-1} dr^2 + r^2 d\Omega_3^2, \quad (2)$$

in which $d\Omega_3^2$ denotes the metric on the unit three-sphere, $d\Omega_3^2 = d\Psi_3^2 + \sin^2\Psi_3 d\Omega_2^2$, are those obtained by Giddings and Strominger [7], and Myers [8] ($n=4$), Hawking [9], and Halliwell and Laflamme [10] ($n=2$), and González-Díaz [11], and Coleman and Lee [12] ($n=1$). These can be readily transformed into Friedmann–Robertson–Walker metrics by introducing the transformation

$$r \rightarrow a = M \cosh^{2/n}(\tfrac{1}{2}n\eta), \quad (3)$$

expressed in terms of the conformal time $\eta = \int d\tau/a$. Again, metrics (2) are euclidean only in the region $r > M$. Hence, in order to draw any sensible parallels between the black hole and wormhole solutions, one should analytically extend metrics (2) into $r < M$, looking then for the region in which such metrics become euclidean.

We shall show in this paper that the maximally extended spherically symmetric metric corresponding to an $n=1$ wormhole of radius M is the unique wormhole metric which, for $r < M$, is conformally equivalent to the maximal inextendible Kruskal metric associated with a Schwarzschild black hole of mass $m = \frac{1}{2}M$. We start by noting that the use of (3) allows us to write

$$da^2 = \left(1 - \frac{M^n}{a^n}\right) d\tau^2. \quad (4)$$

Then, for a given constant value of $\sin\Psi_3 = \epsilon_0$, metric (2) becomes formally the same as metric (1) provided that

$$\frac{\cosh^{2/n}(\tfrac{1}{2}n\eta) - 1}{\cosh^{2/n}(\tfrac{1}{2}n\eta)} + \frac{\sinh^2(\tfrac{1}{2}n\eta) \cosh^{2/n-2}(\tfrac{1}{2}n\eta)}{\cosh^{2/n}(\tfrac{1}{2}n\eta) - 1} = \epsilon_0^{-2} = \text{const.} \quad (5)$$

Condition (5) is satisfied only for $n=2$ (i.e., for the Tolman–Hawking wormhole [9,10]) when $\epsilon_0^2 = \frac{1}{2}$ and $M=2m$. Of course, this is not a true four-dimensional euclidean Schwarzschild metric simply because a and τ are not independent variables, but they are related to each another by the Tolman–Hawking condition $a^2 = M^2 + \tau^2$, so that eq. (4) transforms the obtained metric into either a three-dimensional Robertson–Walker metric or the three-dimensional spherically symmetric metric corresponding to (2) for $n=2$ on a surface of constant angular coordinate $\Psi_3 = \frac{1}{4}\pi$ or $\frac{3}{4}\pi$.

In order to show that it is only the maximal analytical extension of the $n=1$ wormhole which, somewhere, corresponds to the Schwarzschild–Kruskal spacetime [13], let us proceed as follows. Write first the $n=1$ wormhole metric in the form

$$ds^2 = \sin^2\Psi_3 \left[\left(1 - \frac{M}{a}\right)^{-1} da^2 - d\tau^2 + a^2 d\Omega_2^2 \right]. \quad (6)$$

Metric (6) will be defined for a given periodic time interval. Redefine then the Robertson–Walker time τ so that metric (6) formally becomes a Schwarzschild-like metric in terms of the new time variable τ' and $a > M$. Time τ' should then cover the complete time interval from $+\infty$ to $-\infty$. For the still extendible resulting metric to represent a true classical black hole spacetime somewhere, it will be finally re-

quired that (i) the used time transformation must give rise to a two-surface for constant values of new coordinates which should be related to the new time variable τ' and the scale factor a in exactly the same way as constant values of Kruskal coordinates labelling null trajectories in a Schwarzschild spacetime [14] relate to the radial coordinate and Schwarzschild time, in the region $r > 2m$; (ii) such a metric be maximally extended to cover the complete range of the radial coordinate in such a way that the resulting (extended) respective values of Ψ_3 and η are independent of each other.

Metric (6) is obtained for $n=1$ when the angular variable Ψ_3 is identified with the lorentzian conformal time, so that $\Psi_3 = i\eta$, which confines the lorentzian conformal time to the interval $(0, \pi)$. The overall factor in metric (6) becomes

$$\sin^2 \Psi_3 = 4 \frac{a}{M} \left(1 - \frac{a}{M}\right). \quad (7)$$

If it is possible for the $n=1$ wormhole manifold to be maximally extended, $a \rightarrow r$, so that the apparent singularity at $r=M$ of metrics (2) is no longer present, the quantity $\sin \Psi_3$ may then become a real, positive, continuous, and sufficiently differentiable function for $r < M$.

The new time variable can be defined by

$$dt' = \pm \left(1 - \frac{M}{a}\right)^{-1/2} d\tau, \quad (8)$$

for lorentzian metric ($a > M$), or

$$d\tau' = \pm i \left(1 - \frac{M}{a}\right)^{-1/2} d\tau, \quad (9)$$

for euclidean metric ($a > M$). The transformed times t' and τ' cover then the entire time interval. Integration of (8) and (9) produces

$$\pm t' = a + 2M \ln \left(\frac{a}{M} - 1\right)^{1/2} \pm K_1^\pm, \quad (10)$$

for lorentzian time, or

$$\pm i\tau' = a + 2M \ln \left(\frac{a}{M} - 1\right)^{1/2} \pm K_2^\pm, \quad (11)$$

for euclidean time. The K s are arbitrary integration constants which in fact represent two-surfaces [note that from (10) and (11) follows that $-(1-M/$

$a) d\tau'^2 = (1-M/a)^{-1} da^2]$ for constant values of new coordinates which are related to t' (or τ') and a in the required Schwarzschild–Kruskalian form, i.e. defining null trajectories in a spacetime which, after analytical extension, will have exactly the same classical causal structure as the Schwarzschild–Kruskal spacetime in the region $r \leq M$, with $M=2m$. However, considering the radial coordinate as given by the scale factor according to (3) implies a cutting out of the surface at $r=M$, so that the original manifold is divided into two disconnected components. The euclidean description is provided in the submanifold defined in the region $M < a < \infty$, while lorentzian description corresponds to the submanifold defined in the region $0 < a < M$. The first of these submanifolds is then extendible, $a \rightarrow r$, into a new manifold which covers the entire region $0 < r < \infty$ by simply letting t' or τ' to be again independent of the new radial variable r . Thus, the maximal extension of the manifold, $a \rightarrow r$, into the region $0 < r < \infty$, actually represents an increase of the number of coordinates again up to four, and can be implemented by simply replacing the constants K by variables, i.e., the set of coordinates for the maximally extended $n=1$ wormhole manifold is formed by the two angular coordinates of the two-sphere and the two new coordinates. Any transformation of the so-promoted new coordinates would not modify the light-cone structure of the spacetime, provided that the transformation function is real, differentiable and monotonic [15]. Choosing then for the constants $\exp(-K^+/2M) \rightarrow U$, $\exp(K^-/2M) \rightarrow V$, we have

$$U = \left(\frac{r}{M} - 1\right)^{1/2} \exp\left(\frac{\tau}{2M}\right) \exp\left(-\frac{\tau'}{2M}\right),$$

$$V = \left(\frac{r}{M} - 1\right)^{1/2} \exp\left(\frac{\tau}{2M}\right) \exp\left(\frac{\tau'}{2M}\right)$$

(in the region $r > M$) and

$$U = \left(1 - \frac{r}{M}\right)^{1/2} \exp\left(\frac{\tau}{2M}\right) \exp\left(-\frac{\tau'}{2M}\right),$$

$$V = \left(1 - \frac{r}{M}\right)^{1/2} \exp\left(\frac{\tau}{2M}\right) \exp\left(\frac{\tau'}{2M}\right)$$

(in the region $r < M$) for the lorentzian manifold, or the same expressions with τ' replaced by $i\tau'$ for the euclidean manifold.

For the resulting metric to still represent a complete set of geodesics in the $n=1$ wormhole space-time, one should recover independence between conformal time and angular variable Ψ_3 . This is actually the case as can be checked by dividing the two sides of e.g. eq. (8) by a , and integrating the resulting expression. Under the extension $a \rightarrow r$, η' and Ψ_3 should transform into different quantities (say, α' and Ψ), and the involved integration constants should be consistently replaced by dimensionless variables given by

$$\delta = i \sin(\tfrac{1}{2}\Psi) \exp(-\tfrac{1}{2}\alpha'),$$

$$\rho = i \sin(\tfrac{1}{2}\Psi) \exp(\tfrac{1}{2}\alpha').$$

It follows that the transformed conformal time α' and angular variable Ψ are again independent of each other in the maximally extended manifold.

The maximal inextendible $n=1$ wormhole metric becomes then

$$ds^2 = 4 \frac{r}{M} \left(1 - \frac{r}{M}\right) \times \left(\frac{4M^3 dU dV \exp(-\tau/M)}{r} + r^2 d\Omega_2^2 \right). \quad (12)$$

The overall factor for the Schwarzschild-Kruskal metric in (12) is negative for $r > M$. However, when such a factor is defined in terms of the coordinates U and V for $r < M$, it becomes a genuine conformal factor in the interval $0 < r < M$:

$$\Omega^2 = 4UV \exp\left(-\frac{\tau}{M}\right) \left[1 - UV \exp\left(-\frac{\tau}{M}\right)\right], \quad (13)$$

as the function Ω is real, continuous, finite and non-vanishing [16,17] along that interval.

It is not difficult to see that none of the other two possible wormhole metrics in eqs. (2) and (3) could be reducible to a section of the Schwarzschild-Kruskal manifold and, therefore, there is no way for such metrics to be analytically extendible to the Kruskal black hole metric. For example, in the case of an $n=2$ wormhole, the maximal extension of the manifold in the region $M < a < \infty$ to the region $0 < r < \infty$ produces new kruskalian coordinates of the form

$$\left(\frac{r/M-1}{r/M+1}\right)^{1/2} \exp\left(\frac{\tau}{M} \mp \frac{\tau'}{M}\right),$$

whose causal structure (i.e. null geodesics and Cauchy surfaces) is always different of that for the maximally extended metric of any black hole. The same conclusion can be drawn off in the case of the $n=4$ wormhole.

Thus, the main result of this paper is that the maximally extended metric of an $n=1$ wormhole of radius M is the *unique* wormhole metric with a baby universe submanifold, M_{bu} , which can be conformally mapped into the internal black hole submanifold, M_{ibh} , describing the causal structure of the region inside the event horizon of the maximal inextendible Kruskal metric [13] corresponding to a Schwarzschild black hole of mass $m = \frac{1}{2}M$.

Once we have established the existence and uniqueness of a conformal equivalence between the region beyond the event horizon of a black hole and the baby universe spacetime associated with an $n=1$ wormhole, we turn to a brief discussion of the physical consequences. We first notice that, since the larger the neck radius of the $n=2, 4$ wormholes, M , the smaller the nucleation rate per Planck volume per Planck time, $P \sim \exp(-\text{action})$, it appears that this radius can by no means play the role of the radius of a black hole; for this identification would violate the second law of black hole thermodynamics. Nevertheless, in the case of an $n=1$ wormhole, it turns out [6,11] that the larger M , the larger P , so that the surface area of the $n=1$ baby universes could well play the role of entropy, without any violation of the second law. This would also allow a nonnegligible occurrence of macroscopic black holes whose formation and evolution is due to the branching of a closed universe of the size of the black hole.

The classical causal structure of submanifolds M_{bu} and M_{ibh} is the same. Nevertheless, since $r=0$ is a boundary ideal point [18] for M_{ibh} , one should cut it out from M_{ibh} . Hence, for this submanifold to be connected, one should disregard the component beyond $r=0$ [i.e., all those points taking place at negative values of the time τ appearing in eq. (6)]. Now, as $r=0$ is just an ordinary point [18] for M_{bu} , and metrics (6) and (12) are invariant under time reflection, it follows that, whereas each point $q \in M_{ibh}$ represents a single event occurring at $\tau_q > 0$, each point $p \in M_{bu}$ should actually represent a couple of classically indistinguishable events which take place at times τ_p and $-\tau_p$. This would reflect the essential fea-

ture that a black hole can only occur in *one* universe (i.e., a classically disconnected large region of space-time) while a wormhole topology change ought necessarily to occur simultaneously in at least *two*.

The conformal equivalence expressed by (12) for $r < M$ can then be consistently interpreted by considering that the interior of a black hole is not but what is physically manifested out from the occurrence of an $n=1$ wormhole topology change, in only one of the two universes which are just connected by such a topology change.

Metric (12) is defined for an *ideal* observer able to observe events taking place in the two universes simultaneously. Thus, if we would conformally delete all the events taking place at zero and negative values of time τ from metric (12) [i.e., making metric (12) defined for a *real* observer able only to observe events in just one universe, for which there is a singularity at $r=0$], according to the above discussion, the conformal factor would then reduce to unity^{#1}, leaving just the maximally extended metric for a Schwarzschild black hole.

Conformal invariance is broken when one allows a quantum field to propagate through the conformally related spacetimes [17]. It follows that a real observer in our universe would detect a Hawking thermal radiation, characterized by an expectation value of the stress-energy tensor $\langle T_{00} \rangle_{bh} = (129\pi M^2)^{-1}$, coming from a spacetime region where an $n=1$ baby universe of radius M is being branched off, but which behaves like though it were a Schwarzschild black hole of mass $\frac{1}{2}M$ for that observer. A corresponding ideal observer, however, should be expected to detect nothing, for, otherwise, the baby universe would not be closed. In fact, by using the moving mirror approach to particle creation [17,19,20], for an $n=1$ wormhole trajectory

$$z(\tau) = \frac{1}{2}M \left[\left(\frac{\tau^2}{M^2} + 1 \right)^{1/2} + 1 \right] \quad (14)$$

^{#1} Although we have not succeeded in proving this rigorously, the procedure to be followed should preserve the classical causal structure of metric (12) for $0 < r < M$ and involve the introduction of an ideal boundary point at $r=0$. This can be typically achieved by conformally transforming (12). An obvious choice for the corresponding conformal factor is in fact $\sin^{-1}\psi_3$.

[i.e., the $n=1$ scale factor in (3) expressed in terms of the Robertson-Walker time τ], one obtains $\langle T_{00} \rangle_{wh} = 0$.

Then, if the energy is to be conserved, the value of $\langle T_{00} \rangle$ for an ideal observer should be the same as the sum of the $\langle T_{00} \rangle$ for all possible real observers which can be accommodated in the whole event domain of the manifold for the ideal observer. For the case being considered, besides the real observer for M_{bh} , there will be another possible real observer for $\tilde{M}_{bh}$, the negative τ -time copy of M_{bh} . Energy conservation would require then that such an observer will detect a particle flux corresponding to a stress-energy equal to $-\langle T_{00} \rangle_{bh}$. This ultimately implies that, for observers at the other end of an $n=1$ wormhole there must be another black hole which formed from the collapse of a star made up of antimatter, and evaporated giving off the antiparticles to the particles forming the thermal radiation emitted by the first black hole in our region, just as it was conjectured by Hawking [3]. A rather crucial point in the above picture is that, whereas for real observers of M_{ibh} and $\tilde{M}_{ibh}$ there will be the customary black hole singularity, such a singularity will no longer be present for the ideal observer of M_{bu} .

For an observer in our universe, associated to an $n=1$ wormhole of radius M , there will be then an entropy $S_{wm} = \pi M^2$ and a temperature $T_{wm} = 1/4\pi M$, just like for a black hole of mass $\frac{1}{2}M$, and it would radiate according to the Hawking process, shrinking its size until it became of the order of the Planck scale.

This work was carried out while I was together with my mother Nieves in her last days. I wish to thank her inspiration and dedicate the paper to her memory.

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